

# Customer Shopping Behavior Analysis

## 1. Project Overview

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This project analyzes customer shopping behavior using transactional data to identify purchasing trends, customer segments, product performance, and factors influencing repeat purchases. The objective is to convert raw customer data into actionable business insights that support marketing decisions, customer retention strategies, and sales optimization.

## 2. Dataset Summary

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- Rows: 3,900
- Columns: 18
- Dataset includes:
  - **Customer Information:** Age, Gender, Location, Subscription Status
  - **Transaction Details:** Product Category, Purchase Amount, Season, Size, Color
  - **Customer Behavior:** Discounts, Previous Purchases, Purchase Frequency, Reviews, Shipping Preferences

## 3. Data Preparation & Transformation using Excel Power Query

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The raw customer shopping dataset was imported into Excel and transformed using Power Query to prepare a clean and structured dataset for analysis. The data preparation process focused on improving data quality, maintaining consistency, and creating business-ready information for SQL analysis and reporting.

### Data Profiling & Quality Check

Initial data profiling was performed to understand the structure and quality of the dataset. Activities included:

- Reviewed column data types and formats.
- Checked missing and inconsistent records.
- Validated categorical values.
- Analyzed column distribution and data quality.

### Data Cleaning

Data cleaning operations were performed to ensure accuracy and reliability. Key transformations:

- Handled missing values in customer review ratings.
- Removed unnecessary fields.
- Standardized column names and formats.
- Corrected data types for numerical and categorical fields.

## Feature Engineering

Additional analytical fields were created to support deeper business analysis:

- **Age Group:** Customers were categorized into different age segments to analyze purchasing behavior across age groups.
- **Purchase Frequency Days:** Purchase frequency values were converted into numerical day intervals for better comparison and analysis.
- **Customer Segment:** Customers were classified based on purchase history to identify new, returning, and loyal customers.

## Database Preparation

The final transformed dataset was exported and loaded into MySQL for structured querying and business analysis.

## 4. Data Analysis using MySQL

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After preparing the dataset, the cleaned customer transaction data was imported into MySQL for structured analysis. SQL queries were developed to answer key business questions related to revenue performance, customer behavior, product preferences, discount strategy, and customer loyalty. The analysis focused on answering the following business questions:

### 1. Revenue Analysis by Gender

#### Business Question

What is the total revenue generated by male and female customers?

#### Objective

Compare revenue contribution across customer groups and identify differences in purchasing behavior.

#### Result

| gender | revenue |
|--------|---------|
| Male   | 155438  |
| Female | 75191   |

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## 2. High-Value Discount Customer Analysis

### Business Question

Which customers used discounts but still spent more than the average purchase amount?

### Objective

Identify valuable customers who respond to promotions while maintaining high purchase value.

### Result

| Customer ID | Purchase Amount |
|-------------|-----------------|
| 2           | 64              |
| 3           | 73              |
| 4           | 90              |
| 7           | 85              |
| 9           | 97              |
| 12          | 68              |
| 13          | 72              |
| 16          | 81              |
| 20          | 90              |
| 22          | 62              |
| 24          | 88              |
| 29          | 94              |
| 32          | 79              |
| 33          | 67              |
| 1673        | 73              |
| 1674        | 62              |
| 1676        | 90              |

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## 3. Top Rated Product Analysis

### Business Question

Which are the top 5 products with the highest average customer review ratings?

**Objective**

Identify products with strong customer satisfaction and preference.

**Result**

| Item Purchased | Average Product Rating |
|----------------|------------------------|
| Gloves         | 3.86                   |
| Sandals        | 3.84                   |
| Boots          | 3.82                   |
| Hat            | 3.8                    |
| Skirt          | 3.79                   |

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**4. Shipping Preference Analysis****Business Question**

How does the average purchase amount differ between Standard and Express shipping users?

**Objective**

Understand the relationship between shipping preferences and customer spending.

**Result**

| Shipping Type | avg_purchase_amount |
|---------------|---------------------|
| Express       | 60.34               |
| Standard      | 58.41               |

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**5. Subscription Impact Analysis****Business Question**

Do subscribed customers spend more compared to non-subscribed customers?

**Objective**

Evaluate the impact of subscriptions on average spending and total revenue.

**Result**

| Subscription Status | total_customers | avg_spend | total_revenue |
|---------------------|-----------------|-----------|---------------|
| Yes                 | 1033            | 59.42     | 61377         |
| No                  | 2830            | 59.81     | 169252        |

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**6. Discount Dependency Analysis****Business Question**

Which products have the highest percentage of purchases with discounts applied?

**Objective**

Identify products that rely heavily on discounts and promotional strategies.

**Result**

| Item Purchased | discount_rate |
|----------------|---------------|
| Hat            | 49.67         |
| Sneakers       | 49.66         |
| Coat           | 48.43         |
| Sweater        | 47.20         |
| Pants          | 46.43         |

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**7. Customer Segmentation Analysis****Business Question**

How can customers be categorized based on their previous purchase behavior?

**Objective**

Segment customers into New, Returning, and Loyal groups to support targeted marketing.

**Result**

| customer_segment | Number of Customers |
|------------------|---------------------|
| Loyal            | 3085                |
| Returning        | 695                 |
| New              | 83                  |

## 8. Product Performance within Categories

### Business Question

What are the top 3 most purchased products within each product category?

### Objective

Identify best-performing products and understand category-level demand.

### Result

| item_rank | Category    | Item Purchased | total_orders |
|-----------|-------------|----------------|--------------|
| 1         | Accessories | Jewelry        | 170          |
| 2         | Accessories | Sunglasses     | 159          |
| 3         | Accessories | Belt           | 159          |
| 1         | Clothing    | Blouse         | 169          |
| 2         | Clothing    | Shirt          | 168          |
| 3         | Clothing    | Pants          | 168          |
| 1         | Footwear    | Sandals        | 159          |
| 2         | Footwear    | Shoes          | 147          |
| 3         | Footwear    | Sneakers       | 145          |
| 1         | Outerwear   | Jacket         | 163          |
| 2         | Outerwear   | Coat           | 159          |

## 9. Repeat Buyer & Subscription Analysis

### Business Question

Are repeat customers more likely to subscribe?

**Objective**

Analyze the relationship between customer loyalty and subscription adoption.

**Result**

| Subscription Status | repeat_buyers |
|---------------------|---------------|
| Yes                 | 939           |
| No                  | 2503          |

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**10. Revenue Contribution by Age Group****Business Question**

Which age groups contribute the highest revenue?

**Objective**

Identify valuable customer demographics for targeted business strategies.

**Result**

| Age Group   | total_revenue |
|-------------|---------------|
| Middle_aged | 66939         |
| Adult       | 64473         |
| Young Adult | 56669         |
| Senior      | 42548         |

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The SQL analysis results were further used for Power BI dashboard development and business recommendations.

## 5. Dashboard Development using Power BI

An interactive dashboard was built in Power BI to present the key insights from the analysis in a clear, visual, and business-friendly format, enabling stakeholders to explore customer behavior trends at a glance.



## 6. Business Recommendations

Based on the findings from the data analysis, the following strategic recommendations are proposed to support business growth, customer retention, and revenue optimization:

- **Boost Subscriptions** — Promote exclusive benefits for subscribers to increase enrollment and recurring revenue.
- **Customer Loyalty Programs** — Reward repeat buyers and create incentives to move them into the “Loyal” segment.
- **Review Discount Policy** — Balance promotional sales boosts with margin control to protect profitability.
- **Product Positioning** — Highlight top-rated and best-selling products in marketing campaigns to maximize impact.
- **Targeted Marketing** — Focus efforts on high-revenue age groups and express-shipping users to improve campaign ROI.